Johann Benedict Listing — The complete topological writings: *Vorstudien zur Topologie* and *Der Census räumlicher Complexe* (1847/48 + 1862)

The Boneyard is a collection of works left unfinished, unpublished, or unread at their authors' deaths — dry bones in the valley, in Ezekiel's image. Each presentation tells the scholar's story, weighs what survives of his work, and states plainly what completing it would take, so that a book the world never received whole can be met at last. Each of them was one of one. There will never be another like him, and his light will not shine again except through the work he left behind.

A Boneyard dossier — researched and presented by Robert Tilford, Hays, Kansas. Filed 14 October 2026.

Disclosure: prepared with AI assistance — the subject reviewers are AI agents, each desk-checked at this desk; all rulings and final decisions are the author's own.

Dedication

To Johann Benedict Listing (1808–1882), Gauss's student and professor of physics at Göttingen from 1839, who gave a new field of mathematics its name. His essay of 1847 coined the word topology and was an early monograph on the subject, yet he called it only Vorstudien, preliminary studies, announcing a larger work that never appeared. He lectured on topology from 1848 onward, but the Census of 1861/62, with its generalisations of Euler's polyhedron theorem, together with its two addenda, was all more that he ever wrote on it. This edition is dedicated to the man whose word is now in everyday mathematical use, and whose own writings on the subject he named are gathered here whole.

The author's own terms

Listing coined the word topology, explained in this edition as the study of the properties of shapes that remain when a shape is stretched or bent without being torn.

The story

Johann Benedict Listing (1808–1882) gave a new field of mathematics its name, announced in his title that his work was only a beginning, and then never wrote the larger work that the beginning promised.

Listing was Gauss's student, and professor of physics at Göttingen from 1839. His 1847 essay coined the word "topology" and was an early monograph on the subject. Topology is the study of the properties of shapes that remain when a shape is stretched or bent without being torn. That word, now in everyday mathematical use, is Listing's coinage, on the record assembled here.

A stronger claim is sometimes tempting, and this edition does not make it. A "first" for the 1847 essay is UNCHECKED — no hardened check is recorded. Listing is credited here with what the record supports: he coined the word topology, and his essay was an early monograph on the subject.

Even the title tells the unfinished story. The essay is called *Vorstudien*, Preliminary Studies. A title of that kind announces a larger work to come. That larger work never appeared. Listing's follow-up, *Der Census räumlicher Complexe*, published in 1861/62, together with two addenda, was all he ever wrote on topology, although he lectured on it from 1848 onward.

The reviewer who assembled this dossier was candid about the fit: it is weaker on death, and strong on "preliminary / never completed." Listing did not leave his topology unfinished because he died in the middle of it. He left it unfinished because the projected treatise never followed the preliminary studies. That distinction is part of the record, and it is kept here.

The work and what survives

The *Vorstudien zur Topologie* appeared in *Göttinger Studien* 1 (1847), at pages 811–875, and as a booklet published by Vandenhoeck und Ruprecht at Göttingen in 1848. Page images are at HathiTrust, according to the Online Books Page.

Der Census räumlicher Complexe appeared in *Abh. Göttingen* 10 (1861/62), at pages 97–182. It contains generalisations of Euler's polyhedron theorem — that is, early cell complexes, ways of understanding a solid shape through the cells, faces and edges that make it up. In the Edinburgh knot-theory bibliography it is cited in German only.

The German texts of 1847/48 and 1862 are in the public domain.

Its English fortunes

No English translation of either work found after research. English-language histories and bibliographies consulted cite them exclusively in German.

So the man who named topology can be read in English histories of the subject, but his own two writings on it have not been found in English.

What completing it would take

One modest volume would complete the task: the *Vorstudien*, which is short — modern reprints run to about 76–84 pages — together with the *Census* as the author's own sequel. Presented together, they could appear as "Listing on Topology: the complete topological writings."

Such a volume should carry a note that the projected larger treatise implied by "Preliminary Studies" never appeared in the bibliographies searched. That note is not a defect in the collection. It is its point. What survives is the beginning and the sequel Listing actually wrote, complete as a record of his topological work, and honestly marked as less than the work he had announced.

Among the candidates considered in the mathematics report, this is the shortest and cheapest to prepare. Its modesty is its practical virtue: a whole author's writings on a subject he named, gathered in one small book, with no claim made for them beyond what the record supports.

Sources

- https://en.wikipedia.org/wiki/Johann_Benedict_Listing
- https://www.sciencedirect.com/sdfe/pdf/download/eid/3-s2.0-B9780444823755500341/first-page-pdf
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